

Fast Encoding Algorithm for H.265/HEVC Based on Tempo-spatial Correlation

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Abstract

The newest H.265/HEVC video coding standard adopts the quadtree-structured coding unit (CU) to match various texture characteristics of image. Therefore, H.265/HEVC can achieve significantly better coding efficiency than all existing video coding standards such as H.264/AVC. However, the computational complexity of the H.265/HEVC encoder increases dramatically due to its recursive quadtree representation to find the best coding tree unit (CTU) partition. To speed up the encoding procedure, a fast CU depth range decision method is proposed to reduce the depth search range in this paper. Based on the depth information correlation between tempo-spatial adjacent CTUs and the current CTU, some depths can be adaptively excluded from the depth search process in advance. The best depth of current CTU is determined by an intersection between temporal predicted depth ranges by 9 Gaussian weighting from encoded blocks and spatial predicted depth ranges by 4 best weighting from encoded blocks, separately. Experimental results show that the proposed scheme can achieve time improving ratio (TIR) about 42%~82% when compared to the default encoding scheme in HM 8.1 with insignificant loss of image quality. On the other hand, when compared with the other fast algorithm in [8], the proposed algorithm can further achieve an average TIR about 12%~8%.

Keywords: video coding, H.265/HEVC, Gaussian distribution

1. Introduction

With the rapid development of electronic technology, the panels of 4K×2K (or 8K×4K) high-resolution will become the main specification of large size digital TV in future. However, the currently state-of-the-art video coding standard H.264/AVC is difficult to support the video applications of high definition (HD) and ultrahigh definition (UHD) resolution [1]. Therefore, the ITU-T Video Coding

Experts Group (VCEG) and ISO/IEC Moving Pictures Expert Group (MPEG) through their Joint Collaborative Team on Video Coding (JCT-VC) has been developed a newest high efficiency video coding (HEVC) for video compression standard to satisfy the UHD requirement in 2010 [2], and the first version of HEVC was approved as ITU-T H.265 and ISO/IEC 23008-2 by JCT-VC in Jan. 2013 [3-5]. The second version of H.265/HEVC was completed and approved as an ITU-T standard in Oct. 2014 [6].

H.265/HEVC can achieve an average bit rate decrease of 50% in comparison with H.264/AVC High Profile (HP) while still maintaining the same subjective video quality [7]. This is because H.265/HEVC adopts some new coding structures including coding unit (CU), prediction unit (PU) and transform unit (TU) as shown in Figure 1. The CU is the basic unit of region splitting used for inter/intra prediction, which allows recursive subdividing into four equally sized blocks. The CU can be split by coding quadtree structure of 4 level depths, which CU size ranges from largest CU (LCU: 64×64) to the smallest CU (SCU: 8×8) pixels. At each depth level (CU size), H.265/HEVC performs motion estimation (ME) and motion compensation (MC) with different size. The PU is the basic unit used for carrying the information related to the prediction processes, and the TU can be split by residual quadtree (RQT) at maximally 3 level depths which vary from 32×32 to 4×4 pixels. The relationships of quadtree structure depth levels among the CU, PU and TU is shown in Figure 2.

In general, intra CUs have two types of PUs including 2N×2N partition and N×N partition, but inter CUs have four types of PUs including 2N×2N, 2N×N, N×2N and N×N. Therefore, H.265/HEVC encoder enables 7 different partition modes.

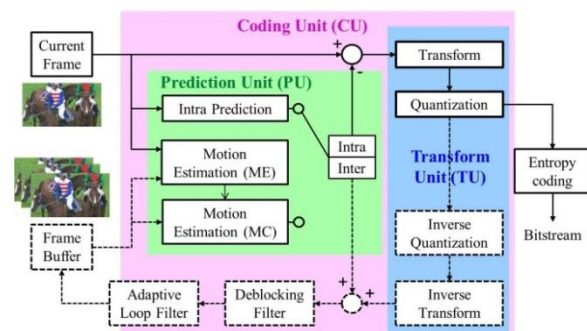


Figure 1: The block diagram of H.265/HEVC encoder.

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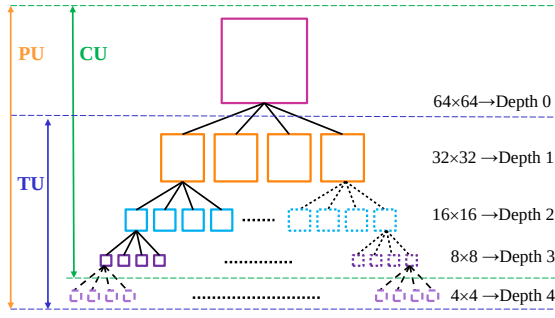


Figure 2: The relationship among the CU, PU and TU.

including SKIP, inter $2N \times 2N$, inter $2N \times N$, inter $N \times 2N$, inter $N \times N$, intra $2N \times 2N$ and intra $N \times N$ for inter slice as shown in Figure 3. The rate distortion (RD) cost under all partition mode and all CU sizes has to be calculated by performing the PUs and TUs to select the optimal CU size and partition mode. For an LCU, all the PUs and available TUs listed in Table 1 have to be exhaustively searched by rate-distortion optimization (RDO) process and this causes dramatically increased computational complexity compared with H.264/AVC. This “try all and select the best” method will result in the high computational complexity and limit the use of H.265/HEVC encoders in real-time applications.

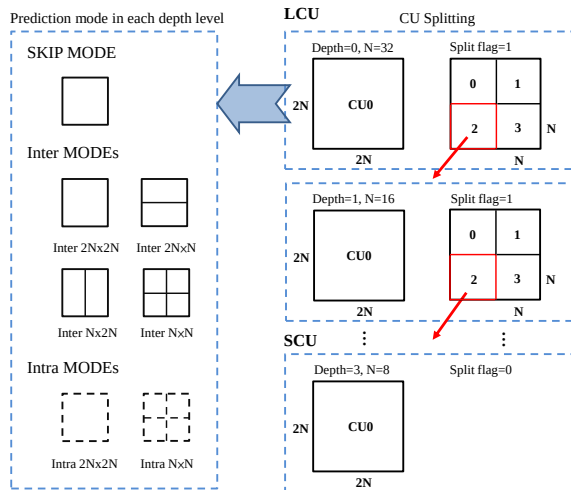


Figure 3: Recursive CU structure, PU splitting for skip, inter and intra modes.

Table 1: PU and TU sizes in H.265/HEVC.

Depth	Inter PU	Intra PU	TU
0	64×64 ,	64×64	32×32 ,
1	64×32 , 32×64	32×32	16×16 ,
	32×32 ,		32×32 ,
	32×16 , 16×32		16×16 ,
2	16×16 , 16×8 ,	16×16	8×8 ,
	8×16		16×16 ,
3	8×8 , 8×4 , 4×8	8×8 , 4×4	8×8 , 4×4

In order to speed up the encoding process of H.265/HEVC, there have been many fast encoding methods proposed to reduce the encoding complexity based on the test model HM 8.1 [8-12]. A simple tree-pruning method has been proposed in coding tree unit (CTU) depth decision by exploiting the SKIP mode and has been accepted by HEVC [3]. The main idea of the scheme is to prune the CTU subtree when its root CU is coded as SKIP mode. This method is simple and effective. But when the SKIP CUs are on the last level of quad tree or not existed, this scheme becomes useless. Another tree-pruning method adopted the Bayesian decision rule to prune CU subtree [9]. This method needs to define threshold by training the data offline. So the threshold accuracy for the untrained video is a problem. All of these are using the tree-pruning technology to accelerate the CTU depth decision process and the difference among these methods is the different pruning condition.

Recently, an adaptive CTU depth range determination (ACUDRD) algorithm is proposed in [8] based on the maximal and minimal values of depth levels to determine current CTU depth. However, the ACUDRD is ineffective for video sequences with active motion or rich texture due to only adopt co-located block to predict the CTU depth range. Therefore, in order to further improve the performance of ACUDRD algorithm, we propose a fast CTU depth range decision algorithm which utilizes the characteristics of strong temporal and spatial correlation in natural video sequences. Firstly, we analyze the temporal correlation and find that the probability is near to Gaussian distribution. And then, the depth of current CTU is predicted by 9 Gaussian weighting from neighboring blocks of the co-located block. On the other hand, we also analyze the spatial correlation and predict the depth of current CTU using 4 best weighting from encoded neighboring blocks. Finally, we determine the best predicted depth range of current CTU from the intersection of two predicted depth range obtained from tempo-spatial adjacent CTU.

The rest of the paper is organized as follows. Section 2 gives some observations and statistical analysis in test model HM. Section 3 we describe the proposed fast encoding method for H.265/HEVC encoder. The results of experiment are shown in Section 4. Finally, Section 5 shows the conclusions of this study.

2. Observations and Statistical Analysis in HM

We start with motivating observations which provide a useful guideline for modelling the correlation between spatio-temporal neighboring CTUs and the current CTU. The test model HM usually adopts the LCU size equal to 64×64 , and the depth level range is from 0 to 3. We first analyze the depth level distribution based on coding of sequences with the different resolutions, and 5 sequences with different motion activities are selected to test the depth level distribution. Test conditions are listed as follows: each test sequence is encoded using the hierarchical B frame structure; quantization parameter (QP) is set to 22, 27, 32 and 36, respectively; RDO is enabled; search range of ME is set to 64. Table 2 shows the optimal depth level distribution in inter slices. Figure 4 shows that an example of the depth level distribution of CTUs between two successive frames.



(a)



(b)

Figure 4: Examples of the depth level distribution of CTUs between two successive frames (QP=32). (a) frame t-1(b) frame t

Table 2: Statistical analysis of depth level distribution.

Sequences	Picture Size	Depth 0	Depth 1	Depth 2	Depth 3
Traffic	2560×1600	33%	14%	27%	26%
Kimono	1920×1080	17%	30%	36%	17%
BQTerrace	1920×1080	25%	9%	23%	43%
BasketballDrill	832×480	18%	16%	32%	34%
Vidyo1	1280×720	62%	12%	16%	10%

Table 2 tabulates that the probability of depth level distribution for the depth level “0”, “1”, “2”, and “3”, respectively. From the statistical results in Table 3, we can find that the sequences containing a large area with high activities such as “Kimono” and “Basketball”, the possibility of selecting depth level “0” is very low with 17% and 18%, respectively. For sequences containing a large area with low activities such as “Vidyo1”, more than 62% CTUs select the depth level “0” as the optimal levels. These results show that small depth levels are always selected at CTUs in the homogeneous region, and large depth levels are selected at CTUs with complex texture. The depth range should be adaptively determined based on the CTU property. On the other hand, we can find that most of CTUs choose the last two depth levels especially for high activity and large motion sequences such as “BQTerrace” and “Basketball”. For “Basketball”, 32% and 34% CTUs choose the optimal depth levels after CU size decision with the depth level “2” and the depth level “3”, respectively, and only about 18% CTUs choose the first depth level “0”.

In addition, as the frame rate highly increasing to 60 Hz, the successive two frames have a stronger tempo-spatial correlation. Therefore, the final depth information or split structure of the current CTU is highly related to the neighboring CTUs’ in the previous frame. Figure 4 shows two certain frames of the test sequence encoded using low-delay (LD) configuration in the HM 8.1 reference software. As shown in Figure 4(a), depth levels of the CTUs in the current frame are the same as the depth levels of temporally nine co-located neighboring CTUs of the previous frame. On the other hand, there are also the same as the depth levels of the spatial four neighboring CTUs in the current frame, as shown in Figure 4(b). Thus, when encoding the current frame, the current CTU can be predicted through the 9 split structure of the co-located CTU in the reference frame and the split structure of the 4 spatial neighboring CTUs in the current frame.

3. Proposed Fast Encoding Algorithm

In this section, a fast CU depth range decision algorithm is described in detail, including motivating observations and algorithm procedure. From Figure 4, we can see that there is significant correlation in depth level information among tempo-spatial neighboring CTUs. Thus when encoding the current CTU, we can predict the searching depth levels of the current CTU by using the depth information of tempo-spatial adjacent CTUs. Based on the above consideration, we should select the neighboring CTU candidates with high correlation as the reference CTU candidates. The neighboring CTUs are selected to predict the depth of the the current CTU as shown in Figure 5.

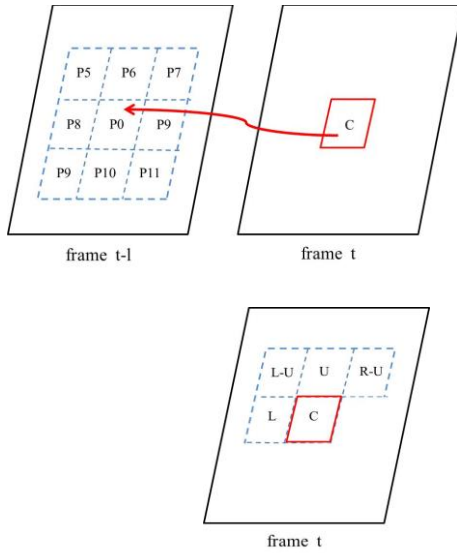


Figure 5: tempo-spatial adjacent CTUs of the current CTU. C: current CTU; L: left CTU; L-U: left upper CTU; U: upper CTU; R-U: right upper CTU; P0: co-located CTU; P5~P12: temporally co-located neighboring CTUs.

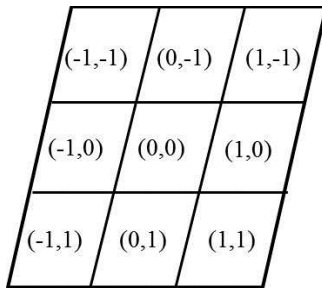


Figure 6: The corresponding coordinate of temporal neighboring CTUs and probability of depth level distribution.

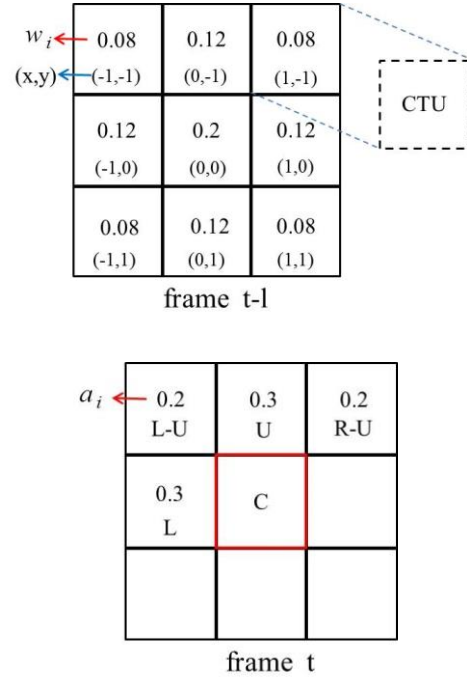


Figure 7: The weights of tempo-spatial correlation value.

To speed up CTU pruning, we make use of tempo-spatial correlations to analyze region properties and skip unnecessary CTU depth levels. The optimal CTU depth level of a block is predicted using temporal and spatial neighbouring blocks, separately. The temporal predicted depth is defined as follows:

$$Depth_{temp_pred} = \sum_{i=0}^{N-1} \omega_i \cdot d_i \quad (1)$$

where N is the number of previous encoded CTUs equal to 9, d_i is the value of depth level and ω_i is the weight determined based on correlations between the current CTU and its temporal neighbouring CTUs as shown in Figure 5. On the other hand, the spatial predicted depth is defined as follows:

$$Depth_{spat_pred} = \sum_{i=0}^{N-1} \alpha_i \cdot d_i \quad (2)$$

where N is the number of previous encoded CTUs equal to 4, d_i is the value of depth level and α_i is the weight determined based on correlations between the current CTU and its spatial neighbouring CTUs as shown in Figure 5. The best predicted depth range of current CTU is determined from the intersection of two predicted depth range as follows:

$$Depth_{best_pred} = Depth_{temp_pred} \cap Depth_{spat_pred} \quad (3)$$

For temporal predicted depth, the nine weights are normalized to get $\sum_{i=0}^{N-1} \omega_i = 1$. Figure 6 shows the corresponding coordinate of temporal neighboring CTUs when the reference coordinate of the co-located CTU is set to (0, 0) and the statistical analysis of the same maximum CTU depth under different sequences. From Figure 6, we can observe that the probability of depth level distribution in the previously coded frame approximates the Gaussian distribution. Therefore, the nine weights are normalized by two-dimension Gaussian function as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (4)$$

where the expected value equal to zero and standard deviation equal to 1 ($\sigma = 1$). In the same statistical way, we can find that the probability distribution in the spatial neighbouring CTUs in the horizontal directions have higher correction of the current CTU than the vertical directions. Figure 7 shows that the weights of spatio-temporal correlation values are set in our proposed method.

According to the predicted value of the best CTU depth, each block is divided into five types as follows:

- 1). If $Depth_{best_pred} = 0$, its best CTU depth is chosen to "0". The dynamic depth range (DR) of current CTU is classified as Type 0.
- 2). If $0 < Depth_{best_pred} \leq 0.5$, its best CTU depth is chosen to "0". The dynamic DR of current CTU is classified as Type 1.
- 3). If $0.5 < Depth_{best_pred} \leq 1.5$, its best CTU depth is chosen to "1". The dynamic DR of current CTU is classified as Type 2.
- 4). If $1.5 < Depth_{best_pred} \leq 2.5$, its best CTU depth is chosen to "2". The dynamic DR of current CTU is classified as Type 3.
- 5). If $Depth_{best_pred} > 2.5$, its best CTU depth is chosen to "3". The dynamic DR of current CTU is classified as Type 4.

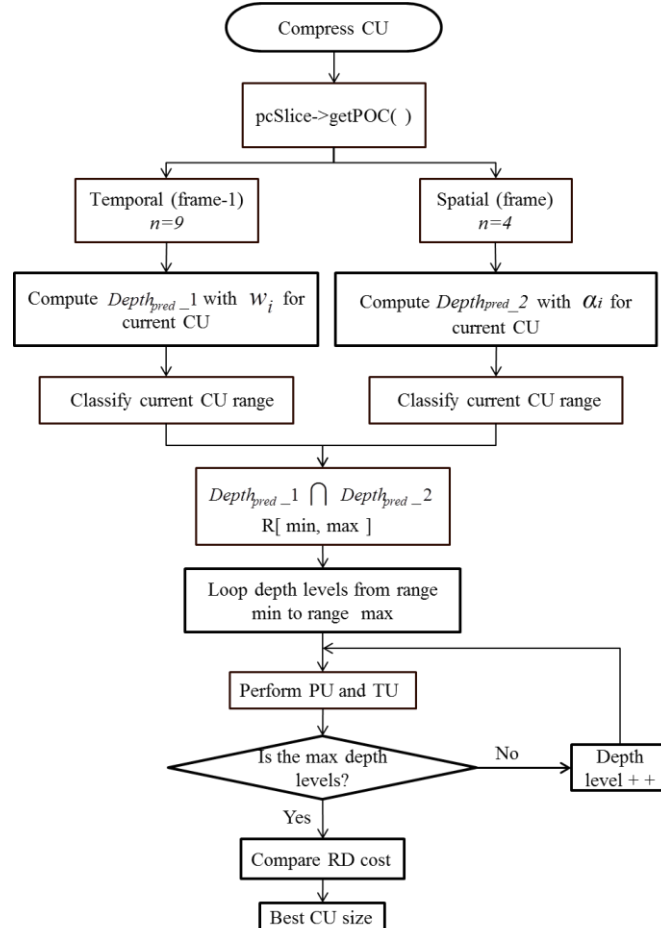


Figure 8: Flowchart of the proposed fast encoding algorithm in HM 8.1.

Table 3: Candidate depth levels for each CTU type.

Types	$Depth_{best_pred}$	Depth Range [$Depth_{min}$, $Depth_{max}$]
Type 0	$Depth_{best_pred} = 0$	[0 , 0]
Type 1	$0 < Depth_{best_pred} \leq 0.5$	[0 , 1]
Type 2	$0.5 < Depth_{best_pred} \leq 1.5$	[0 , 2]
Type 3	$1.5 < Depth_{best_pred} \leq 2.5$	[1 , 3]
Type 4	$Depth_{best_pred} > 2.5$	[2 , 3]

Based on the above analysis, the candidate depth levels that will be tested using RDO for each CTU are summarized in Table 3. Therefore, the dynamic depth level range of CTU of the current CTU can be predicted by using the correlation weight and maximum depth levels of its neighboring blocks. The flowchart of proposed fast encoding method for H.265/HEVC encoder implemented in HM 8.1 [15] is shown in Figure 8. The proposed algorithm firstly utilizes 9 temporally co-located neighboring CTUs to predicted depth level of current CTU. And then, it follows to calculate the predicted depth level value using 4 spatial neighboring CTUs. Finally, we determine the best predicted depth range of current CTU from the intersection of two predicted depth range obtained from tempo-spatial adjacent CTU.

4. Simulation Results

For the performance evaluation, we assess the total execution time of the proposed method in comparison to those of the HM 8.1 [14-15] and the ACUDRD in order to confirm the reduction in computational complexity. The coding performance is evaluated based on Δ Bitrate, Δ PSNR and Δ Time, which are defined as follows:

$$\Delta \text{Bitrate} = \frac{\text{Bitrate}_{\text{method}} - \text{Bitrate}_{\text{HM8.1}}}{\text{Bitrate}_{\text{HM8.1}}} \times 100\% \quad (5)$$

Table 4: Test conditions and software reference configurations.

Test sequences	• Class A (2560×1600): Traffic • Class B (1920×1080): Kimono, ParkScene, Cactus, BasketballDrive, BQTerrace • Class C (832×480): BasketballDrill, BQMall, PartyScene
Total frames	100 frames
QP	22, 27, 32 and 37
Software	HM 8.1
Scenario Type	Main profile (MP) Low delay P (IPPP...), Random access (IBBB...)

$$\Delta \text{PSNR} = \text{PSNR}_{\text{method}} - \text{PSNR}_{\text{HM8.1}} \quad (6)$$

$$\Delta \text{Time} = \frac{\text{TIME}_{\text{HM8.1}} - \text{TIME}_{\text{method}}}{\text{TIME}_{\text{HM8.1}}} \times 100\% \quad (7)$$

The system hardware is Intel (R) Core(TW) CPU i7-3350P @ 3.40 GHz, 8.0 GB memory, and Window XP 64-bit O/S. Additional details of the encoding environment are described in Table 4.

To evaluate the performance of the proposed method, we incorporated it into test model HM8.1 and implemented it using test sequences recommended by JCT-VC in different resolutions including 2560×1600 (Traffic), 1920×1080 (Kimono, ParkScene, Cactus, BasketballDrive, BQTerrace) and 832×480 (BasketballDrill, BQMall, PartyScene). For a fair comparison, we also implemented the ACUDRD algorithm in [8] in same encoding environment. Table 5 and Table 6 tabulate the performances obtained by testing the HM8.1, the ACUDRD and the proposed method with QP=32, separately. As shown in the Table 5, in a high-efficiency setting with the random access (RA) and the low delay (LD) scenarios, the CTU processing time of proposed method can achieve TIR about 55.73% for RA and 65.77% for LD on average, respectively, as compared to the HM8.1 with insignificant loss of rate distortion (RD) performance which includes bitrate and image quality. On the other hand, Table 6 also shows the performances between the ACUDRD and the proposed method with the same scenario. We also can find that our method can further achieve an average of TIR to 18.33% and 12.31% with RA and LD scenarios, respectively, as compared with the ACUDRD algorithm. It is clear that the proposed fast encoding method indeed efficiently reduce the computational complexity in CU module with insignificant loss of RD performance.

Table 5: Comparisons between HM and proposed method.

Sequence	RA			LD		
	HM8.1/ Proposed			HM8.1/ Proposed		
	Δ Bitrate (%)	Δ PSNRY (dB)	TIR (%)	Δ Bitrate (%)	Δ PSNRY (dB)	TIR (%)
Traffic	1.67	-0.12	-42.32	-	-	-
Kimono	0.55	-0.01	-71.23	1.09	-0.01	-76.23
ParkScene	0.77	-0.08	-56.29	1.39	-0.03	-69.21
Cactus	2.47	-0.06	-55.36	1.75	-0.02	-56.22
BasketballDrive	2.35	-0.04	-66.68	0.98	-0.02	-80.00
BQTerrace	-0.08	-0.05	-53.88	-1.18	-0.07	-63.99
BasketballDrill	1.61	-0.06	-55.70	2.09	-0.04	-82.49
BQMall	2.12	-0.11	-48.77	4.37	-0.06	-52.90
PartyScene	0.64	-0.04	-51.34	0.92	-0.01	-46.13
Average	1.34	-0.06	-55.73	1.43	-0.03	-65.77

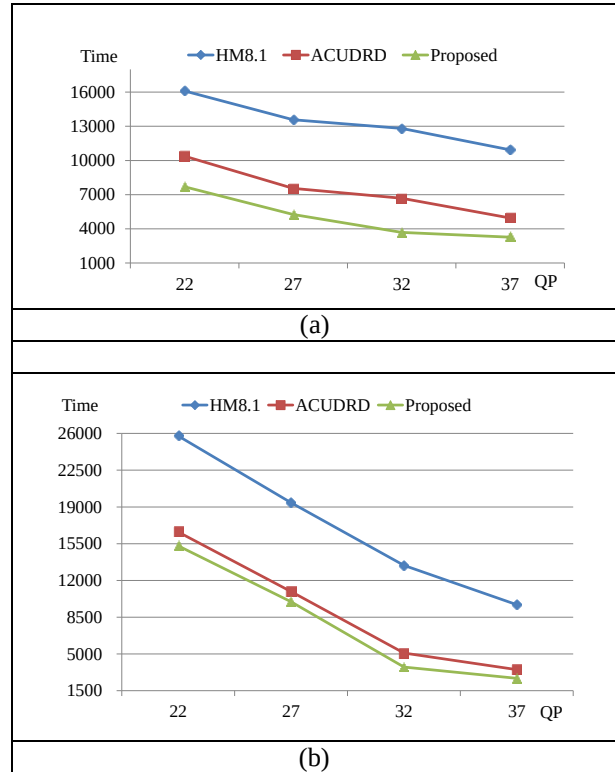
Table 6: Comparisons between ACUDRD and proposed method.

Sequence	RA			LD		
	ACUDRD / Proposed			ACUDRD / Proposed		
	Δ Bitrate (%)	Δ PSNRY (dB)	TIR (%)	Δ Bitrate (%)	Δ PSNRY (dB)	TIR (%)
Traffic	0.86	-0.08	-2.16	-	-	-
Kimono	0.13	0.00	-44.79	0.77	0.01	-17.70
ParkScene	0.07	-0.05	-19.42	0.67	-0.01	-13.28
Cactus	1.24	-0.04	-15.41	0.82	-0.01	-10.32
BasketballDrive	1.23	-0.02	-19.15	0.46	-0.01	-10.76
BQTerrace	-0.06	-0.04	-1.62	-0.96	-0.06	-26.07
BasketballDrill	0.72	-0.02	-15.03	0.70	-0.01	-6.26
BQMall	0.81	-0.09	-23.63	2.56	-0.01	-8.40
PartyScene	0.21	-0.03	-23.76	0.51	0.00	-5.68
Average	0.58	-0.04	-18.33	0.69	-0.01	-12.31

Figures 9(a) and 9(b) show that the CU module processes time for Kimono and BQTerrace sequences tested by different methods with different QP values, respectively. As shown in Figure 9, the proposed method can improve the CU encoding time under different QPs. However, the encoding time is close to that of ACUDRD when the value of QP increases. This is because the quantization error is too large that results in the lower tempo-spatial correlation.

5. Conclusion

We propose a fast CU depth range decision method by exploiting the characteristics of high tempo-spatial correlation exists in nature video sequences. The simulation results show that the proposed algorithm can achieve an average TIR about 42%~71% for RA and 46%~83% for LD, respectively, when compared to H.265/HEVC (HM8.1). Compared with the ACUDRD algorithm [8], the proposed algorithm can further achieve an average TIR about 2%~45% for RA profile and 6%~26% for LD profile, respectively. It is clear that the proposed fast encoding algorithm can efficiently reduce the computational complexity of H.265/HEVC encoder with insignificant loss in terms of image quality and bitrate.

**Figure 9: The CU module processing time with different QP (a) Kimono (b) BQTerrace**

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